Reg. No. [ ] [18EE53]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

POWER ELECTRONICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	Explain the different types of power electronic circuits with neat diagrams.	10
	b.	Compare the operating features of MOSFET with that of IGBT.	04
2.	a.	With the help of two transistor model, derive an expression for anode current of thyristor.	08
	b.	Discuss the need of protection against di/dt ad dv/dt and also explain how it is achieved.	06
3.	a.	Draw the circuit of resonant pulse commutation and explain its function with the relevant waveforms and also derive its expression for circuit turn off time.	10
	b.	Compare Forced commutation and natural commutation	04

OR

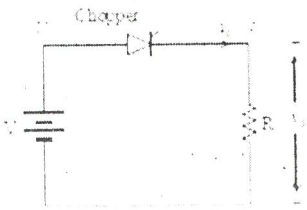
4.	a.	Discuss the principle of ON-OFF control in AC voltage controller and also mentions its two applications	08
	b.	An antiparallel connection of SCR & diode is connected in series with resistive load of 6Ω. The circuit is supplied with 230 V, 50 Hz source. The firing angle of SCR is π/2. Calculate i. Average output voltage ii. RMS output voltage iii. Input power factor	06
5.	a.	A Single phase half wave controlled rectifier is used to supply power to 10Ω load from 230 V, 50 Hz supply at a firing angle of 30°. Calculate i. Average Output Voltage ii. Effective Output Voltage iii. Average Load Current iv. Rectification Ratio v. Form Factor	08
	b.	With neat waveforms explain the working of single phase full bridge converter with RL load and also obtain the expressions of average and rms voltage.	06

OR

6. a. With necessary circuit diagram and waveforms, explain the operation of three phase half wave converter. **07**

b. The SCRs are triggered at a delay angle of 60°. The input voltage is 440 V, 3 Phase, 50 Hz. Write the expression for average DC voltage and RMS output voltage of 6 pulse converter and also find the average DC voltage and RMS output voltage available at the bridge terminals. **07**

7. a. A dc chopper in figure has a resistive load of R = 10Ω and input voltage of V = 200 V. When chopper is ON, its voltage drop is 2 V and the chopping frequency is 1 kHz. If the duty cycle is 60%, **08**



Compute

i. Average output voltage

ii. RMS value of output voltage

iii. Effective input resistance of chopper

iv. Chopper efficiency.

b. Explain the operation of step down chopper for continuous load current with relevant waveforms and also derive the expression for average output voltage. **06**

OR

8. a. Explain single phase half bridge inverter with necessary circuit diagram and waveforms. Derive the equation for rms output voltage. **07**

b. Draw the waveforms and explain how voltage is controlled in single phase inverter by using multiple pulse width modulation technique. **07**